

2-step PCR protocol

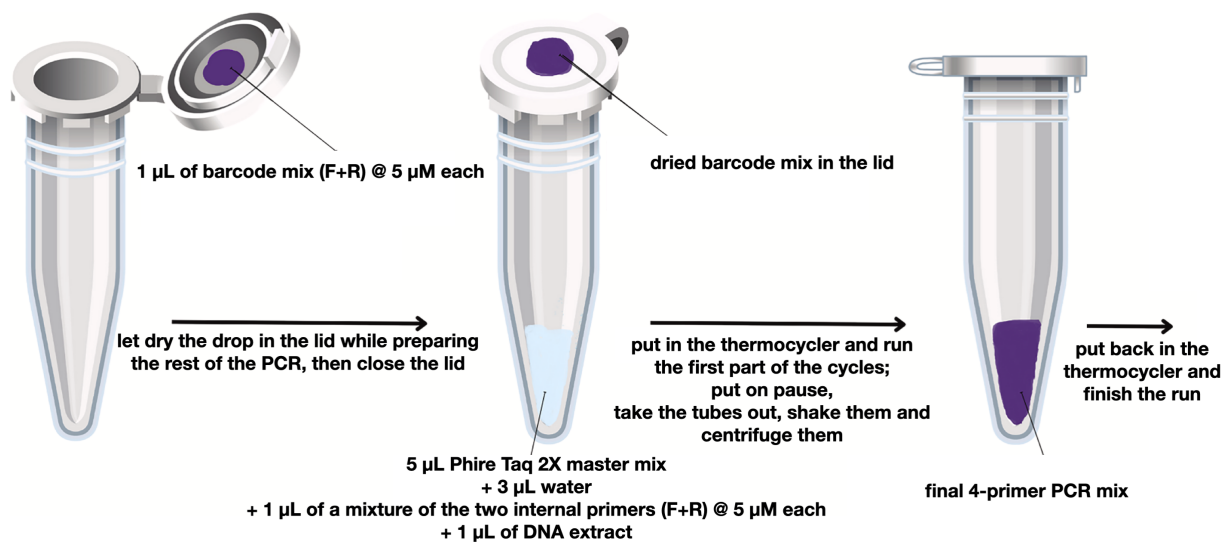
PCR mastermix

In the lid of each tube, place 1 μL of the barcode mix (5 μM final concentration for each), then let it dry under the PCR hood while preparing the rest (below).

Ingredient	Quantity (in μL)	Comment
Phire Taq polymerase MM 2X	6	
Water	4	
Mix of F and R primers (5 μM final concentration for each)	1	Since the PCR is 10 μL in total, the final concentration of each primer in the PCR is 0.5 μM , which is the recommended amount in the Phire Taq protocol.
DNA extract (around 10 ng of DNA)	1	Phire polymerase is very sensitive to concentration. If some samples are not working, it is likely that they are too concentrated

Notes:

- 1) I recommend working using strips of 8 tubes. Since we will need to centrifuge them, it is better to have an even number of strips (e.g. 2x8 or 4x8 PCRs).
- 2) Don't forget to always include both a negative control (adding water instead of a DNA extract) and a positive control (a DNA extract that worked previously) whenever possible.
- 3) Our common dilutions for primer aliquots in the lab is 10 μM . If you mix in equal amount F and R primers at 10 μM each, the resulting mixture contains a final concentration of 5 μM of each primer, which is exactly what we need for the PCR.



note: the barcode mix is represented in blue color for visualization only, but in reality it is transparent

PCR cycle

	Cycle name	Temp (°C)	Time (in second)	Number of cycles	Comment
step 1 (inner primers only)	Initial denaturation	98	300		
	Denaturation	98	8	17	
	Annealing	Depend on your inner primers	8		50°C for 27f-1492r universal bacterial primers; 45°C for the COI Folmer universal primers; 53°C for primers defined using Primer3 with a default Tm of about 60°C
	Extension	72	Depend on the marker length		According to the Phire Taq manual, elongation should be 20s per kb (with a minimum of 20s if the expected length of the PCR product is less than 1 kb). Hence 20s for the COI Folmer fragment (658 bp), 20s for 16S (1500 bp), etc.
	Hold	10	infinite	infinite	Here, the thermocycler will remain on pause. Remove the tubes from the thermocycler, invert them and make the master mix fall into the lid by tapping gently the tubes on the table. Centrifuge the racks of tube in a microcentrifuge to make the mix fall down to the bottom of the tubes, then put them back in the thermocycler and proceed to next steps
Step 2 (with barcodes)	Denaturation	98	8	30	
	Annealing	60	8		50°C for 27f-1492r universal bacterial primers; 45°C for Folmer universal primers; 53°C for primers defined using Primer3 with a default Tm of about 60°C
	Extension	72	Depend on the marker length		According to the Phire Taq manual, elongation should be 20s per kb (with a minimum of 20s if the expected length of the PCR product is less than 1 kb). Hence 20s for the COI Folmer fragment (658 bp), 20s for 16S (1500 bp), etc.
	Hold	10	600		PCRs can then stay at room temperature (keeping them at 10°C for long periods can damage the PCR machines)

Important: never include a “final extension” step. This is a common mistake that can be found in many PCR protocols, but it is very bad as it increases the amount of recombinant sequences among the amplicons.

Gel preparation

Name	Amount
Agarose	1.2 g
TBE 1X	100 ml

- Heat the mix 1-2 minutes in microwave
- Let it cool
- Add 5 µl of Midori
- Pour the gel on an electrophoresis stand and let it solidify
- Put the gel in an electrophoresis cell
- Add 3 µl of each PCR product in each well and let the DNA migrate at 140 volt for 20 minutes

Beads cleanup for Nanopore sequencing

1. Base on the thickness of the band of your gel, pull more or less PCR product per barcode in an Eppendorf tube.
2. Mix well the beads using a vortex.
3. Prepare freshly 500 µl of 80% ethanol
4. Mix your PCR product and the beads in a ratio 1.6:1 (for 100 µl of PCR product add 62.5 µl of beads) and pipette up and down slowly 10 times
5. Comment: this will efficiently remove anything bellow 500-400 bp (the dimers). But you can vary the ratio depending on your target length
6. Incubate at room temperature for 5 minutes
7. Place the tube on the magnetic stand until the supernatant is cleared.
8. Remove the clear liquid from the tube without disturbing the beads. Remove as much as possible
9. With the tube on the magnetic stand, wash the beads by adding 200 µl of freshly prepared 80% ethanol and let it incubate 30 second.
10. Remove the ethanol
11. Repeat step 9 a second time
12. Remove the ethanol and let it dry for around 3-5 minutes. Do not overdry (it should not look cracked)
13. Remove the tube from the magnet and add 21 µl of nuclease free water for elution.
Gently pipette up and down 10 times to resuspend the beads
14. Place the tube back on the magnet until the supernatant is cleared.
15. Carefully transfer 20 µl of supernatant to a new tube.
16. Measure DNA concentration with Qbit